Midterm Report

Start Assignment

- Due Tuesday by 5pm
- Points 15
- Submitting a file upload
- Available after Oct 12 at 2pm

Midterm Report 2: The Bus

Individual assignment. Same rules as Report 1: shared numbers, your own analysis and writing.

Everything here comes from **Part 2** — the edge captures, the two histograms and the capacitances you extracted from them.

15 points. Due: First 60 minutes of the session at the end of week 2, the same deadline as your Part 2 demonstration. Late submission costs 10% per hour.

Question 3: The two buses (10 points)

a. Include your two rise-time histograms on common axes, with mean and standard deviation for each bus.

b. Report the capacitance you extracted for each bus, and the difference.

c. Bus 0 carries three devices and bus 1 carries two. Is the difference you measured consistent with that? Estimate the capacitance one device adds, and comment on whether the pins alone can account for it or whether the trace length matters too.

d. Report the maximum SCL frequency each bus can support under the I²C rise-time limits. Compare with the maximum in the LSM303 datasheet. If those disagree, say which number describes the chip and which describes your board.

e. You now have two independent routes to the same physical limit: the maximum SCL frequency *predicted* from your measured rise time, and the frequency at which your Lab 5 acknowledgment actually stopped appearing. Put both numbers side by side. If they disagree by more than a factor of two, one of them is wrong or something else is limiting the bus — say which you believe and why.

Question 4: How close to the physics are you? (5 points)

Every measurement has a thermodynamic floor. This one is a long way from it.

a. Using the capacitance you measured, compute the kT/C noise voltage on the bus at room temperature.
b. Convert that voltage into timing jitter, using the slew rate of your own measured rising edge at the point it crosses the logic threshold.
c. Compare with the standard deviation of your measured rise times. Which is larger, and by what factor?
d. Name the three effects most likely responsible for the difference, and propose a measurement that would isolate one of them.

> You will find you are several orders of magnitude above the thermal limit. That is the expected answer. Knowing where the floor is, and knowing you are nowhere near it, is more useful than a standard deviation on its own.

Submission

Upload a single PDF, `midterm_report2.pdf`, containing your answers, both histograms and your edge captures. Label the axes and give units.

Your **code goes to the Midterm – Part 2 assignment**, uploaded before your demonstration.